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Mid Term Exam – Technical Report

By

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Abstract

This report presents a comprehensive machine learning analysis of transportation systems using clustering, classification, and regression techniques. The datasets include traffic sensors, environmental, and operational variables characterized by skewness, missing values, outliers, and multicollinearity. A leakage-safe preprocessing pipeline was implemented, including imputation, robust scaling, encoding, and outlier capping to ensure data quality and model reliability.

Clustering analysis identified three stable traffic states as free flow, moderate congestion, and severe congestion with K-Means outperforming hierarchical clustering (silhouette ≈ 0.235 vs 0.215). In classification, the Artificial Neural Network (ANN) achieved the best test performance (accuracy ≈ 0.865, weighted F1 ≈ 0.859), outperforming Random Forest, which exhibited overfitting (train accuracy ≈ 0.986 vs test ≈ 0.850). Hyperparameter tuning further confirmed ANN as the preferred model (cross-validated F1 ≈ 0.902).

In regression, both models performed well, with the MLPRegressor slightly outperforming Gradient Boosting on test data (RMSE ≈ 5.53 vs 5.59, R^2 ≈ 0.917 vs 0.916), while Gradient Boosting provided interpretable feature importance. The analysis revealed that navigation_eta_minutes is the dominant predictor (~90% importance).

In general, the results demonstrate that robust preprocessing combined with appropriate model selection enables accurate and reliable transportation analytics for real-world applications.

Keywords: Transportation analytics, clustering, classification, regression

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1 Introduction

Transportation analytics combines operational data from roadway infrastructure, sensors, weather feeds, and route descriptors to support decision-making in traffic management and intelligent transportation systems. Such data are inherently heterogeneous, they contain a mixture of continuous and categorical variables, non-normal distributions, missing values, correlated measures, and occasionally extreme observations caused either by real traffic disruptions or sensor artifacts. Consequently, effective model development requires not only algorithm selection but also strong upstream data engineering.

The report had two objectives. First, it aimed to build a defensible end-to-end machine learning workflow that mirrors professional transportation analytics setting. Second, it aimed to interpret the results in operational terms rather than merely reporting metrics. The report therefore treats preprocessing as a substantive modeling decision, applies both unsupervised and supervised methods, and evaluates each model using metrics appropriate to its task.

2 Problem 1: Data Understanding and Preprocessing

2.1 Dataset description

At the initial stage, all datasets for clustering, classification, and regression tasks were loaded into the analysis environment, along with their corresponding test sets for subsequent use. The primary objective of this step is to gain a clear understanding of the dataset's structural characteristics (Table 1).

Specifically, the analysis focuses on identifying:

- The dimensionality of each dataset,
- The types of variables present,
- The probable target column in supervised learning tasks, and
- The presence of any potential identifier columns.

This step is critical, as it ensures that all preprocessing decisions are informed by the actual properties of the data rather than based on assumptions.

To support this process, the analysis is conducted on separate datasets for clustering, classification, and regression, each with dedicated training and test partitions. This separation is methodologically important, as it prevents data leakage between exploratory model development and final evaluation. Additionally, schema inspection confirms that all six data files have been successfully loaded and are available for further processing.

Table 1. Overview of the datasets

Dataset	Rows × Columns	Role	Key variables / target
Clustering train	1200 × 12	Unsupervised pattern discovery	hour_of_day, road_type, lane_count, avg_speed_kph, traffic_flow_vehph, occupancy_pct, queue_length_m, precipitation_mm, incident_flag, signal_density_per_km, device_batch
Clustering test	420 × 12	Pattern stability check	same schema as clustering train
Classification train	1800 × 18	Delay-risk prediction	weather_state, visibility_km, signal_density_per_km, avg_speed_kph, traffic_flow_vehph, upstream_flow_vehph, queue_length_m, sensor_quality_score; target = delay_risk_class
Classification test	600 × 18	Generalization check	same schema as classification train
Regression train	1600 × 15	Travel-time prediction	route_length_km, distance_km, weather_state, signal_density_per_km, bus_stop_density_per_km, demand_index, avg_speed_kph, navigation_eta_minutes, driver_experience_level; target = travel_time_minutes
Regression test	520 × 15	Forecast evaluation	same schema as regression train

2.2 Exploratory data analysis (EDA)

EDA was intentionally problem-driven rather than decorative. Histograms, boxplots, count plots, and correlation heatmaps were used to determine whether downstream preprocessing should be robust to skewness, sparsity, and extreme values.

2.2.1 Clustering Dataset

The clustering dataset exhibits strong variability in traffic conditions, with queue length and traffic flow showing clear right-skewness and extreme values, indicating congestion events (Figure 1). Differences in feature scales further highlight the need for normalization. Categorical variables such as road type show moderate imbalance (Figure 2).

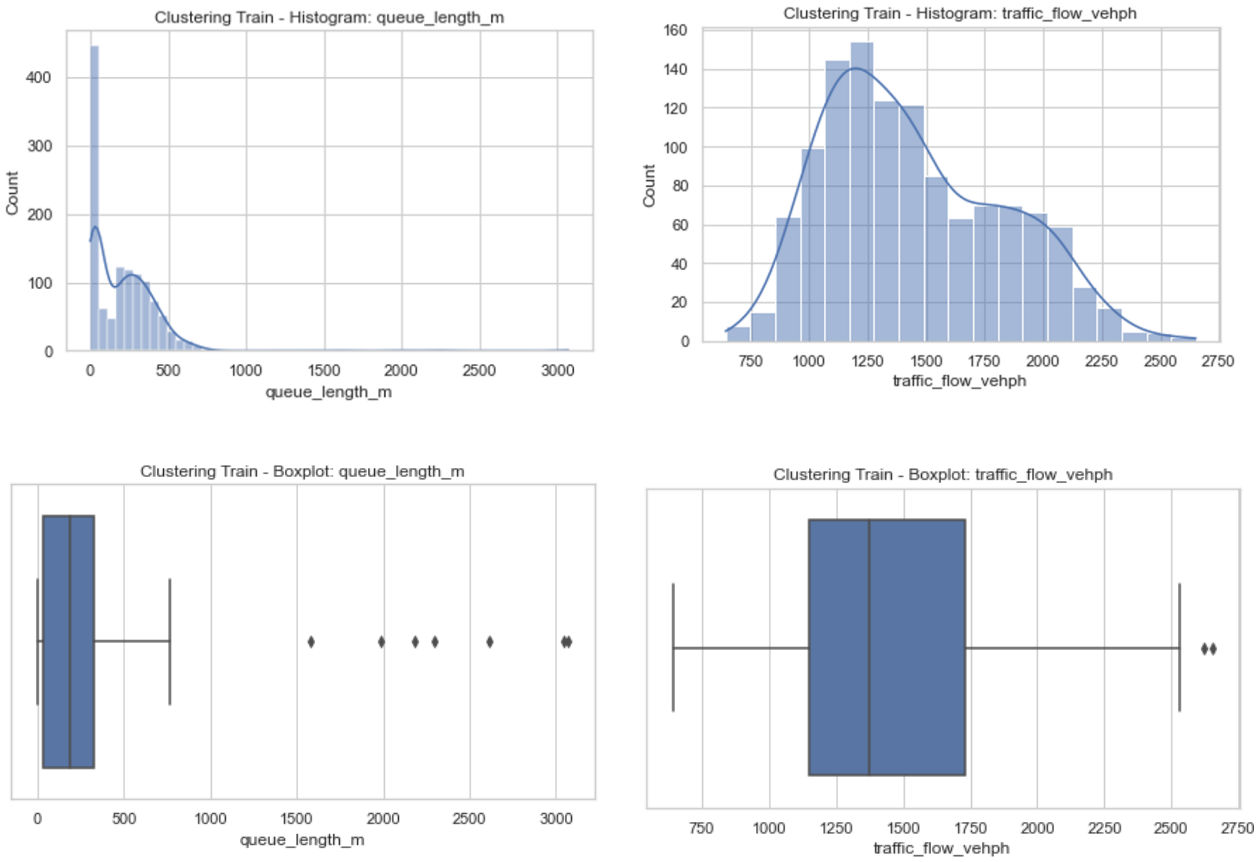


Figure 1. Identification of extreme congestion-related observations and distribution spread

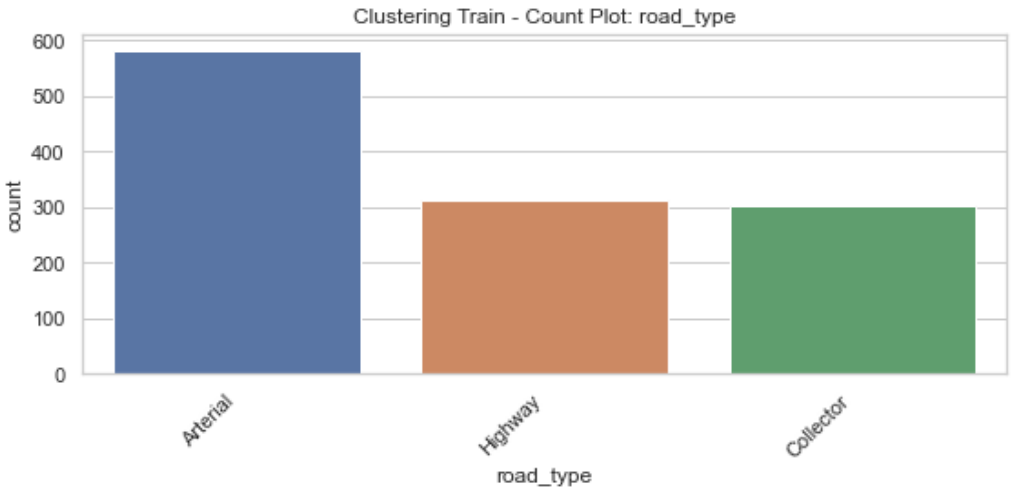


Figure 2. Distribution of road types

2.2.2 Classification Dataset

The classification dataset exhibits significant class imbalance, with the majority labeled as Low delay risk (Figure 3). Traffic flow and queue length show extreme upper values and skewness (Figure 4), while strong dependencies exist between flow-related variables (Figure 6). Environmental features (e.g., weather) also vary unevenly (Figure 5).

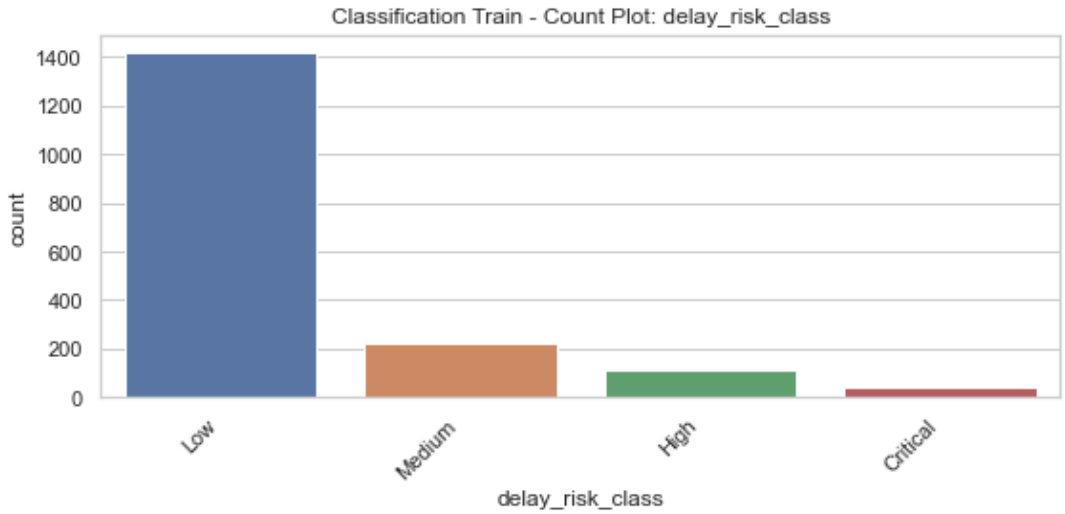


Figure 3. Distribution of target classes showing significant imbalance

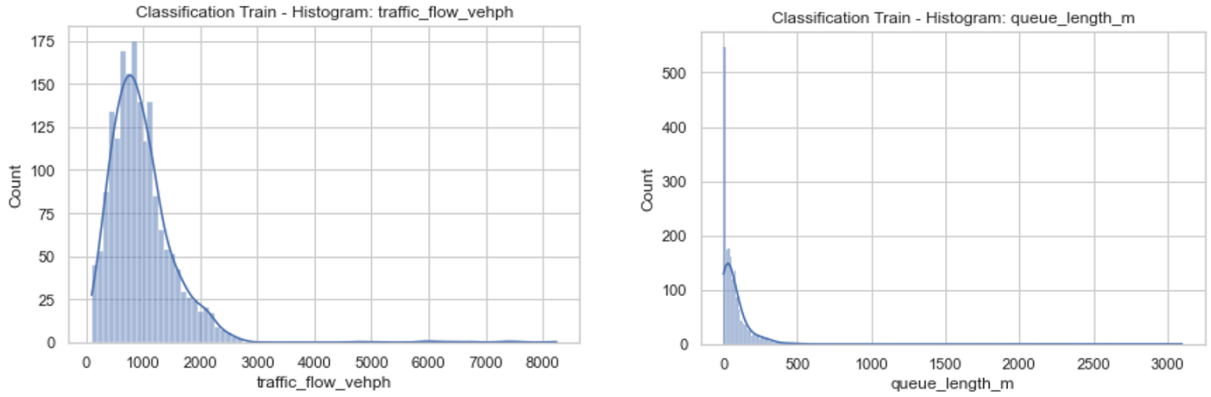


Figure 4. Right-skewed distributions of traffic flow and queue length

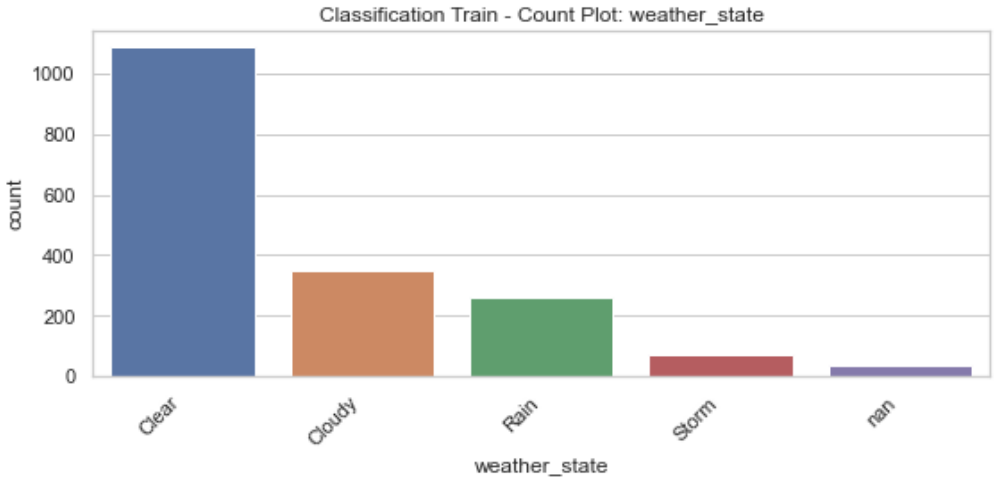


Figure 5. Imbalanced distribution of weather conditions

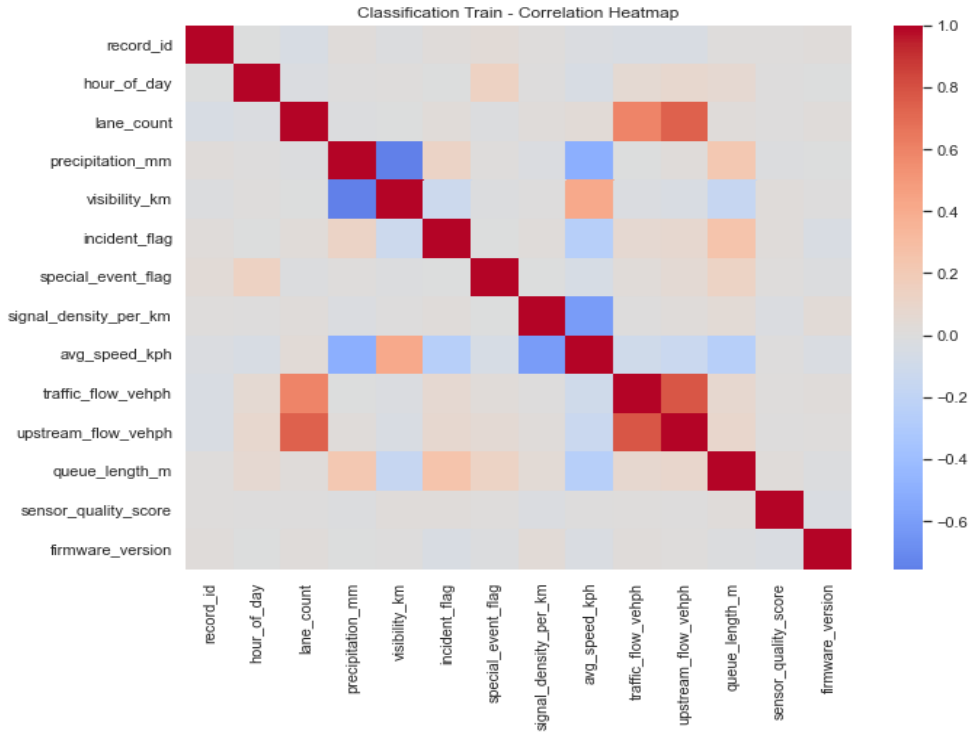


Figure 6. Relationships among traffic variables showing interdependence

2.2.3 Regression Dataset

The regression dataset shows wide variability and right-skewness in travel time and ETA, reflecting congestion effects (Figure 7). Strong relationships are expected between distance, speed, and travel time, while categorical variables such as road type and weather are unevenly distributed (Figure 8).

The heatmap shows strong relationships among key variables, especially between distance, speed, ETA, and travel time, indicating that these factors are closely linked and jointly influence traffic prediction (Figure 9).

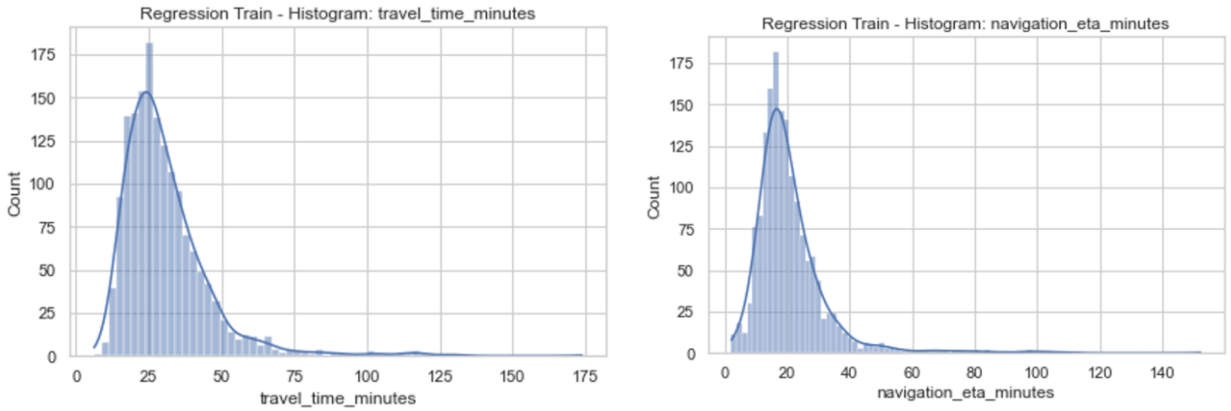


Figure 7. Right-skewed distributions of travel time and ETA

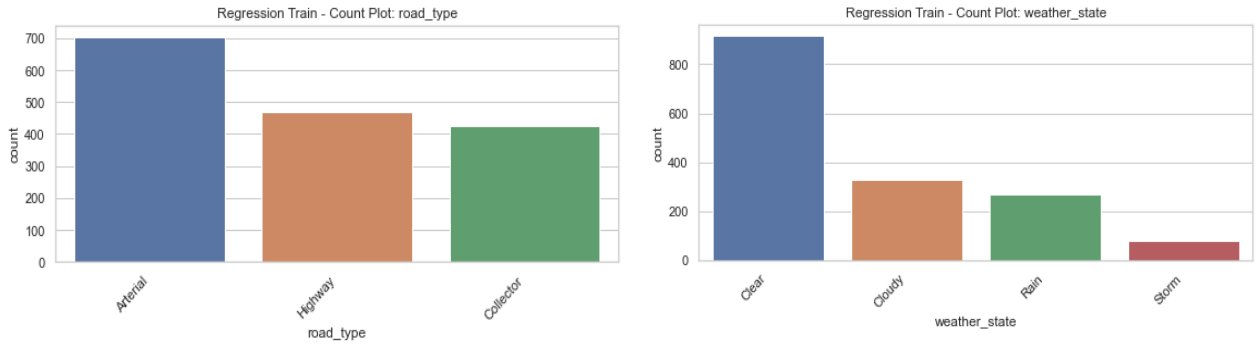


Figure 8. Imbalanced distribution of road type and the environment

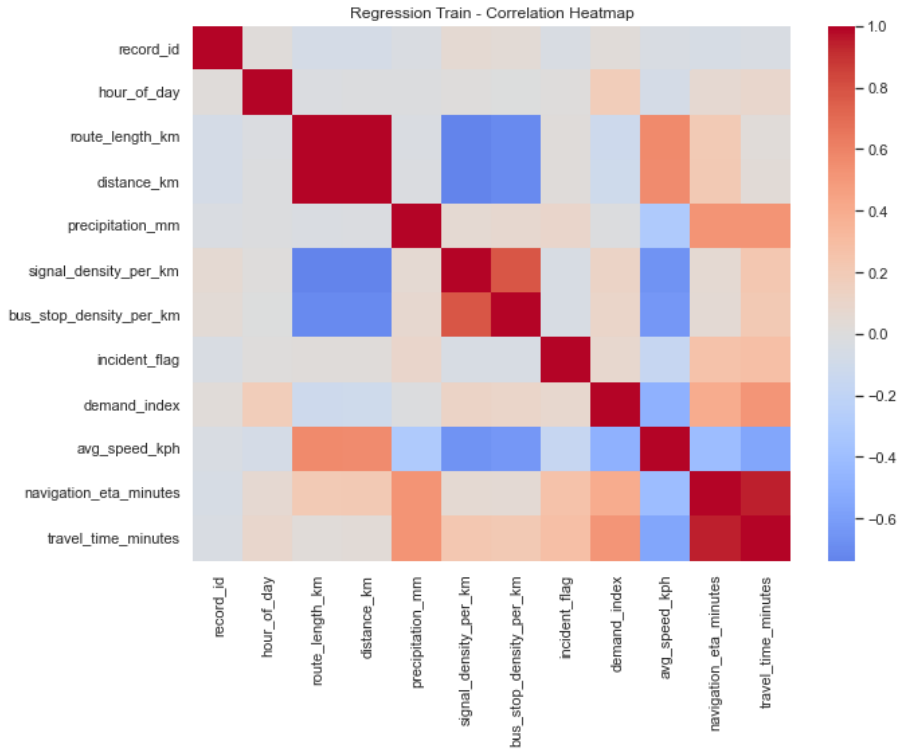


Figure 9. Strong correlations among key variables

In conclusion, the EDA underscores that transportation data is inherently skewed, interdependent, and subject to meaningful extremes. These characteristics necessitate thoughtful preprocessing and domain-aware modeling to ensure reliable and interpretable results.

2.3 Missing values analysis

Missing values are a common characteristic of transportation datasets and arise from several practical factors, including sensor outages, communication failures, manual recording errors, and the absence of contextual information. If left unaddressed, missing data can distort traffic modeling outcomes, reduce the effective sample size, and introduce bias into analytical results.

To address this, missingness was quantified prior to selecting any imputation strategy. This step is essential, as indiscriminate row deletion can remove valid observations and distort the distribution of rare but operationally important traffic states (e.g., congestion events). The analysis identified moderate levels of missingness across all three training datasets (Table 2), but not at a level that would justify record deletion.

Table 2. Summary of variables with missing values

Dataset	Variables with missing data	Missingness
Clustering train	precipitation_mm; avg_speed_kph; occupancy_pct	6.0%; 5.0%; 4.0%
Classification train	visibility_km; avg_speed_kph; sensor_quality_score; weather_state	5.0%; 3.94%; 3.0%; 2.0%
Regression train	precipitation_mm; bus_stop_density_per_km; avg_speed_kph; driver_experience_level	5.0%; 4.0%; 4.0%; 3.0%

These patterns support the use of imputation rather than case-wise deletion. In transportation systems, missing values typically result from temporary issues such as detector outages, communication loss, or incomplete contextual data, rather than fundamentally invalid observations. Given that missingness levels generally range between 2% and 6%, applying appropriate imputation techniques allows for the preservation of sample size while minimizing the introduction of bias or excessive synthetic information.

2.4 Outlier detection

Outliers are a critical aspect of transportation data analysis, as extreme values may represent either genuine rare events such as traffic incidents, road closures, or severe congestion or measurement errors arising from faulty sensors or corrupted records. Distinguishing between these two cases is essential, as inappropriate handling can lead to loss of valuable information or biased modeling outcomes.

To systematically identify outliers, the Interquartile Range (IQR) method was employed. This approach is non-parametric, robust to non-normal distributions, and well suited for transportation variables, which often exhibit skewness and heavy tails. Outliers were further examined using boxplots to visualize their distribution and intensity.

The analysis revealed that outliers are particularly concentrated in precipitation-related variables and incident indicators, followed by key traffic measures such as queue length, average speed, and navigation ETA in supervised datasets. The extent of outliers across datasets is summarized below (Table 3).

Table 3. Extent of outliers across datasets

Dataset	Most affected variables	Outlier rate
Clustering train	precipitation_mm; incident_flag	13.92%; 13.75%
Classification train	precipitation_mm; queue_length_m; avg_speed_kph; incident_flag	18.06%; 7.17%; 6.19%; 5.72%
Regression train	precipitation_mm; incident_flag; navigation_eta_minutes; travel_time_minutes	21.45%; 7.69%; 5.25%; 4.69%

These findings indicate that extreme values are not isolated anomalies but are relatively common in transportation datasets, particularly under adverse weather or incident conditions.

Outliers have a significant impact on modeling performance. They can distort distance-based clustering, destabilize neural network optimization, bias scale-sensitive algorithms, and affect summary statistics used for interpretation. However, given that many of these extreme observations correspond to meaningful real-world events, indiscriminate removal is not appropriate.

Instead, a winsorization (outlier capping) strategy was adopted during preprocessing. This approach limits the influence of extreme values by capping them at defined thresholds, while preserving all observations in the dataset. This ensures that rare but important traffic events remain represented, while reducing their disproportionate impact on model training.

In conclusion, the chosen approach balances statistical robustness with domain relevance, making it well suited for transportation applications where extreme conditions are both frequent and informative.

2.5 Correlation analysis

Transportation datasets inherently contain highly correlated variables, as multiple measurements often capture different aspects of the same underlying traffic state (Table 4). Common examples include relationships between speed and travel time, volume and density, and occupancy and congestion-related indicators.

To assess these relationships, correlation analysis was conducted using a threshold of $|r| > 0.75$ to identify strongly correlated numerical variable pairs. This threshold provides a practical basis for detecting potential multicollinearity and redundancy within the feature space.

Table 4. Extent of outliers across datasets

Dataset	High-correlation pairs (|r| > 0.75)	Interpretation
Clustering train	avg_speed_kph vs occupancy_pct (0.8271)	speed and occupancy describe the same congestion state from different angles
Classification train	traffic_flow_vehph vs upstream_flow_vehph (0.7807); precipitation_mm vs visibility_km (0.7549)	flow propagation and adverse-weather coupling are operationally plausible
Regression train	route_length_km vs distance_km (0.9947); navigation_eta_minutes vs travel_time_minutes (0.9450); signal_density_per_km vs bus_stop_density_per_km (0.7802)	indicate strong structural redundancy, particularly in route characteristics and time-based measures

Multicollinearity is important because it can inflate variable importance, reduce interpretability, decrease efficiency in certain models, and introduce redundancy. This is especially evident in the regression dataset, where route length and distance are nearly interchangeable, and navigation ETA closely aligns with travel time. While these relationships are expected in transportation systems, they require careful interpretation, as models may distribute importance across correlated variables or allow one to dominate.

Despite these findings, correlated variables were not automatically removed. Tree-based models, which are commonly used in such contexts, are generally robust to multicollinearity. Therefore, feature removal is approached conservatively to avoid discarding meaningful signal. Instead, correlation analysis is used as a diagnostic tool to guide interpretation and inform model design decisions.

2.6 Impact of data issues on modeling

The observed data characteristics missing values, outliers, skewness, and multicollinearity have direct and significant consequences for model behavior in transportation applications.

1) Missing values can destabilize estimators or lead to the unnecessary loss of scarce observations if handled through case-wise deletion. Given that missingness often occurs during critical traffic conditions (e.g., sensor outages during congestion), improper handling may reduce model robustness and limit its ability to capture important operational states.
2) Outliers are particularly influential in transportation modeling. While many extreme values represent genuine events such as incidents or severe congestion, they can adversely affect model performance. For example, clustering algorithms such as K-Means rely on Euclidean

distance and can be disproportionately pulled toward extreme observations, leading to distorted cluster structures. Similarly, outliers can disrupt optimization in machine learning models and bias summary statistics.

3) Multicollinearity introduces additional challenges, particularly in model interpretation. Highly correlated variables can inflate the apparent importance of related features, create instability in coefficient estimates, and lead to redundancy in the feature space. This is especially problematic when the focus is on feature importance and interpretability rather than purely predictive accuracy.

Overall, these issues underscore the importance of careful, domain-aware preprocessing. Addressing missingness, controlling outliers, managing skewness, and diagnosing correlation are essential steps to ensure that transportation models remain stable, interpretable, and reflective of real-world traffic dynamics.

2.7 Preprocessing steps and justification

Following the identification of key data issues missing values, skewness, outliers, and multicollinearity a structured preprocessing pipeline was implemented to ensure that the datasets are suitable for modeling while preserving meaningful transportation signals. The preprocessing was applied separately for clustering, classification, and regression tasks to maintain methodological clarity and task-specific relevance.

2.7.1 Identifier Removal

Identifier fields such as record_id were removed using a dedicated transformation step. These variables do not encode any transportation behavior and instead represent row identity. Retaining them would introduce noise and potentially allow models to learn artificial ordering patterns rather than true traffic dynamics. Their removal ensures that only behaviorally relevant features contribute to model learning.

2.7.2 Imputation

Missing values were handled using median imputation for numerical variables and mode imputation for categorical variables. This choice is directly supported by the EDA findings:

- Numerical traffic variables (e.g., speed, flow, queue length) exhibit skewness and outliers, making the median a more robust measure than the mean.

- Categorical variables (e.g., weather_state, device_batch, driver_experience_level) are nominal, and mode imputation preserves valid observed categories without introducing artificial labels.

This approach ensures that missing data is addressed without distorting the underlying distribution or removing valuable observations.

2.7.3 Scaling

Two scaling strategies were considered, with RobustScaler predominantly adopted across workflows:

- RobustScaler was used for most preprocessing pipelines because it relies on median and IQR, making it resistant to outliers and skewed distributions commonly found in transportation data.
- StandardScaler remains conceptually useful in cases where distributions are more stable and where models (e.g., neural networks) benefit from zero-centered features.

The selection of RobustScaler is therefore justified by the presence of extreme values and non-normal distributions in traffic variables.

2.7.4 Encoding

Categorical variables such as road_type, weather_state, zone_type, and driver_experience_level were encoded using OneHotEncoder. This method is appropriate because these variables are nominal and lack inherent ordering.

One-hot encoding prevents the introduction of false ordinal relationships and allows models to learn independent effects for each category. It also ensures compatibility with both tree-based and neural network models.

2.7.5 Outlier Treatment

Outliers were not removed outright. Instead, a winsorization (capping) strategy was applied within preprocessing. This decision reflects the dual nature of outliers in transportation data:

- Extreme values may represent meaningful events such as congestion, incidents, or severe weather.
- At the same time, they can disproportionately influence model behavior, especially in distance-based and scale-sensitive algorithms.

Capping reduces the influence of extreme values while preserving all observations, ensuring that rare but important traffic conditions remain represented in the data set.

2.7.6 Pipeline Application and Outcomes

Reusable preprocessing pipelines were applied separately for each task:

- *Clustering*: Applied directly to the full dataset using RobustScaler.
- *Classification*: Target variable (delay_risk_class) was separated before preprocessing.
- *Regression*: Target variable (travel_time_minutes) was similarly isolated prior to transformation.

This separation ensures task-specific feature spaces, prevents target leakage, and maintains modular, reproducible pipelines. After preprocessing, feature dimensions increased due to one-hot encoding (Clustering: 12 to 15, Classification: 18 to 24, Regression: 15 to 20). All features were scaled, missing values fully imputed, and the datasets are now consistent and ready for model training.

3 Problem 2: Traffic Pattern Discovery Using Clustering

3.1 Introduction

Clustering analysis aims to identify latent traffic states (e.g., free flow, moderate congestion, severe congestion) from the unsupervised dataset. As clustering is an unsupervised task, it is performed directly on the processed data obtained from Problem 1.

The report applies K-Means and hierarchical clustering, determines the optimal number of clusters, interprets each cluster as a traffic state, visualizes the results, and evaluates clustering quality using appropriate metrics.

3.2 Methodology

Clustering analysis was conducted using both K-Means and Hierarchical (Agglomerative) clustering to identify latent traffic patterns in the dataset. These methods were selected to provide complementary perspectives that K-Means offers efficient partitioning based on distance minimization, while hierarchical clustering provides a flexible structure for exploring data relationships through a dendrogram.

3.2.1 K-Means clustering

K-Means clustering was applied to the preprocessed training dataset to identify latent traffic patterns. The optimal number of clusters was determined using both the Elbow Method and the Silhouette Score.

The Elbow plot shows a clear reduction in inertia as the number of clusters increases (Figure 10), with a noticeable bend around $k = 4–5$, after which the rate of improvement diminishes. This indicates that adding more clusters beyond this point yields limited benefit in reducing within-cluster variance.

The Silhouette analysis provides additional insight (Figure 11). The highest score is observed at $k = 2$ (≈ 0.61), suggesting well-separated clusters at this level. However, this may oversimplify the traffic system into only two states. For $k \geq 3$, silhouette scores drop significantly ($\approx 0.20–0.25$), indicating weaker separation but potentially more meaningful granularity.

Considering both metrics and transportation interpretability, $k = 3$ or $k = 4$ is a reasonable choice, as it balances cluster separation with the ability to represent multiple traffic regimes (such as free flow, moderate congestion, and heavy congestion).

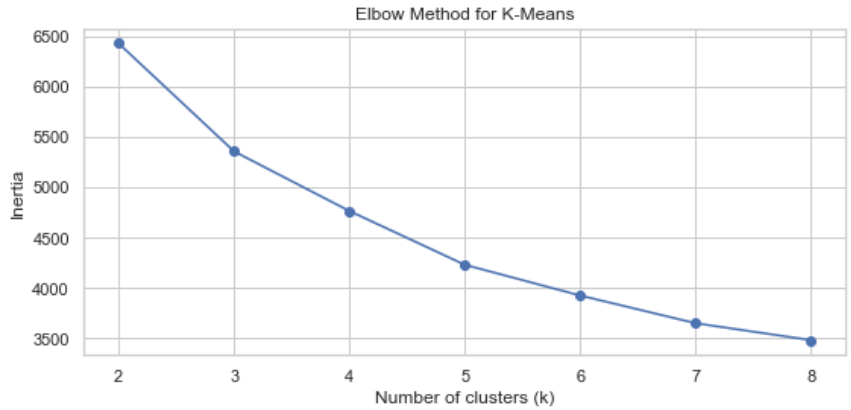


Figure 10. Elbow curve for K-Means

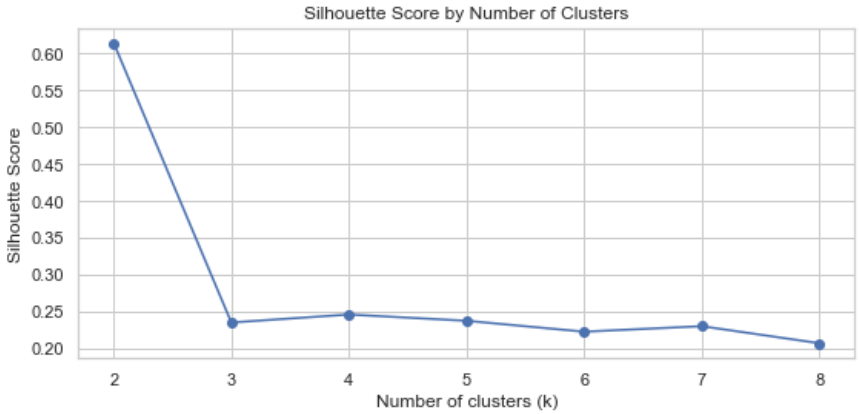


Figure 11. Silhouette score by number of clusters

3.2.2 Hierarchical Clustering

Hierarchical (agglomerative) clustering was performed using Ward linkage, which minimizes within-cluster variance and is consistent with the objectives of K-Means.

The dendrogram reveals a clear hierarchical structure in the data, with large linkage distances appearing at higher levels, indicating natural separations between major clusters (Figure 12). The structure suggests that the data can be meaningfully partitioned into approximately 3 to 4 clusters, consistent with the findings from K-Means.

Unlike K-Means, hierarchical clustering does not assume spherical clusters and provides a more flexible view of data structure. The dendrogram allows visual inspection of cluster formation and supports the selection of a cut level that aligns with practical interpretation.

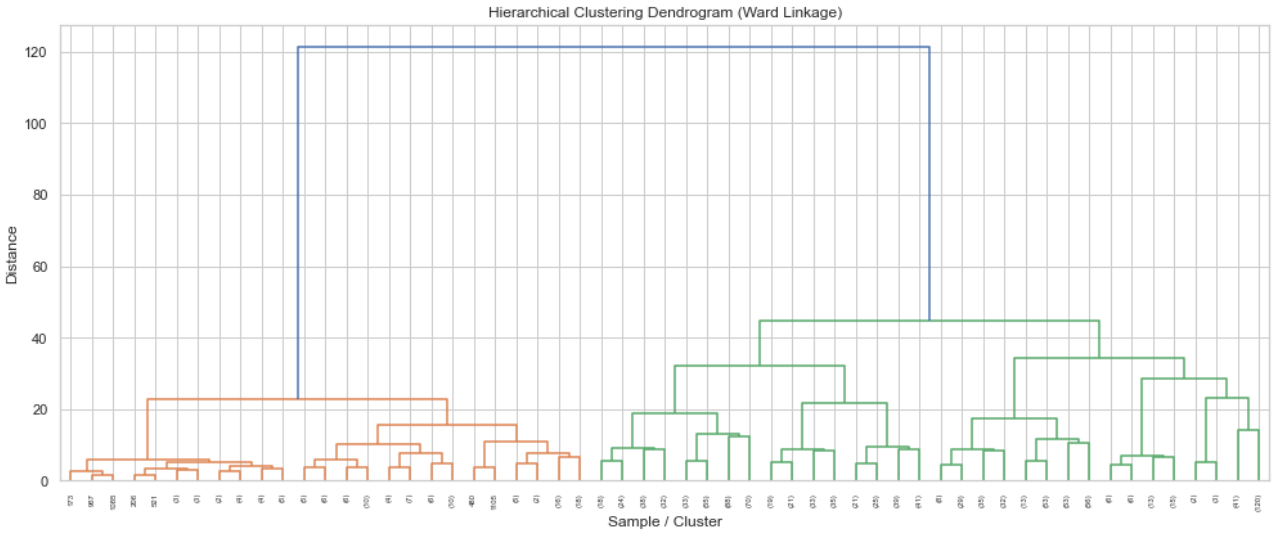


Figure 12. Dendrogram Showing Natural Grouping of Traffic Observations

3.3 Selection of optimal number of clusters

The optimal number of clusters was determined using both the Elbow Method and Silhouette Score. The silhouette score was highest at k=2 (0.614), indicating strong cluster separation. However, k=2 produces only two broad traffic states, which may be insufficient for transportation analysis.

The Elbow Method shows a significant reduction in inertia up to k=3, after which the improvement becomes more gradual. Although the silhouette score decreases at k=3, this value provides a better balance between cluster quality and interpretability.

Additionally, the hierarchical dendrogram indicates natural groupings beyond two clusters, supporting a slightly more granular segmentation. Therefore, k=3 was selected as the optimal number of clusters, as it allows meaningful classification of traffic conditions into free flow, moderate congestion, and heavy congestion.

3.4 Cluster interpretation

The final clustering solution was obtained using K-Means with k = 3, with hierarchical clustering (Ward linkage) applied for comparison. Cluster centers were first analyzed in the transformed feature space, and interpretation was refined by examining average values in the original dataset grouped by cluster.

Cluster profiles reveal clear differences in traffic conditions:

- *Cluster 0 (Free Flow)*: Characterized by high average speed (76.2 kph), low occupancy (19.4%), low queue length (30 m), and minimal incidents. This represents uncongested traffic conditions.
- *Cluster 1 (Severe Congestion)*: Shows very low speed (3.41 kph), high occupancy (72.7%), long queues (354.5 m), and the highest precipitation and incident rates. This reflects heavily congested and disrupted traffic states.
- *Cluster 2 (Moderate Congestion)*: Exhibits moderate speed (34 .2kph), high flow (1648.5 veh/h), elevated occupancy (69.4%), and substantial queue length (326 m), indicating stable but congested conditions.

Cluster labels were assigned based on relative traffic intensity, the lowest-intensity cluster was labeled Free Flow, the intermediate cluster Moderate Congestion, and the highest-intensity cluster Severe Congestion.

This approach ensures a consistent and data-driven interpretation in the absence of ground-truth labels. Overall, the clustering results successfully capture meaningful traffic regimes that align with real-world transportation behavior, supporting their use in downstream monitoring and decision-making applications.

3.5 Visualizations and evaluation

The PCA scatter plot shows that Free Flow is compact and well separated on the left-hand side of the projection, indicating consistent and low-traffic conditions while Severe Congestion is displaced primarily along the first principal component, showing distinct high-congestion conditions. Moderate Congestion occupies an intermediate but broader region, indicating more variability in traffic behavior (Figure 13).

Overall, the clusters are well separated, especially along PC1, confirming that the model successfully distinguishes different traffic states.

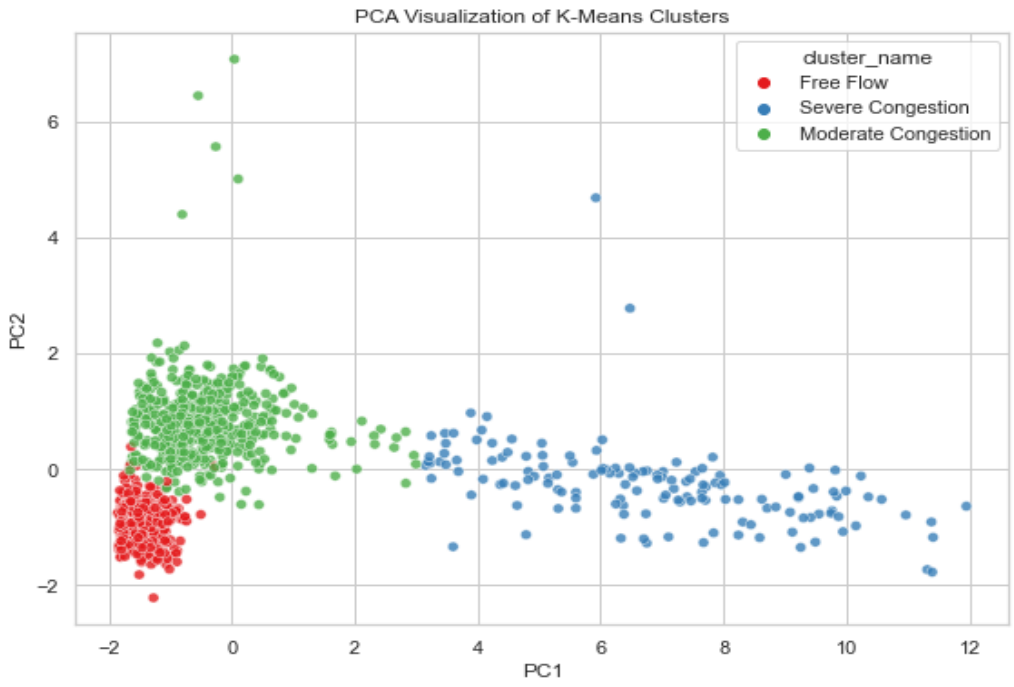


Figure 13. Dendrogram Showing Natural Grouping of Traffic Observations

In metric terms, K-Means outperformed Agglomerative clustering on both reported criteria. Its silhouette score was 0.234988 versus 0.214685, and its Davies-Bouldin index was lower at 1.453919 versus 1.585311 (Table 5). The difference is not enormous, but it is sufficient to prefer K-Means as the final descriptive model.

Table 5. Clustering performance comparison

model	silhouette_score	davies_bouldin_index
K-Means	0.234988	1.453919
Agglomerative	0.214685	1.585311

The trained K-Means model was applied directly to the processed test matrix. The resulting cluster shares are stable between training and test data. The distribution of clusters in the training and test data is very similar. Severe congestion stays almost the same (about 12% in both), while free flow and moderate congestion change only slightly (Figure 14). This shows that the clusters are stable and not just specific to one dataset, meaning the traffic patterns identified are consistent and likely to appear in new data as well.



Figure 14. Comparison between training and test clustering behavior

4 Problem 3: Traffic Condition Classification

4.1 Introduction

Classification analysis aims to predict traffic condition classes (e.g., free flow, moderate traffic, heavy congestion) using traffic sensor data. The models are trained in processed datasets to ensure consistency and data quality.

Two models, Random Forest and Artificial Neural Network (ANN) are trained and evaluated using accuracy, precision, recall, and F1-score. Hyperparameter tuning is applied to improve performance, and the final models are tested on a separate dataset to assess generalization.

4.2 Model Training

The classification target was delay_risk_class, a categorical variable representing the level of traffic delay risk. Two classification models were trained using the processed dataset, a Random Forest classifier and an Artificial Neural Network (MLPClassifier). The use of both models enables a structured comparison between a tree-based ensemble method and a neural network approach.

4.2.1 Random Forest

The Random Forest model was used due to its strong performance on tabular transportation data, ability to handle nonlinear relationships, and robustness to feature interactions and outliers. The configuration was carefully chosen to improve generalization: a higher number of trees (n_estimators = 500) enhances stability, while limiting tree depth (max_depth = 20) and adjusting splitting criteria (min_samples_split = 5, min_samples_leaf = 2) help prevent overfitting. Additionally, max_features = "sqrt" increases model diversity and reduces the impact of correlated variables, which are common in traffic datasets.

4.2.2 Artificial Neural Network

The Artificial Neural Network (MLPClassifier) was used to capture complex nonlinear patterns in the data, such as relationships between flow, speed, and congestion. A deeper architecture (hidden layers: 128, 64, 32) allows the model to learn hierarchical representations. To ensure stable training, a lower learning rate (learning_rate_init = 0.0005) was used, along with ReLU activation and the Adam optimizer, which are well suited for tabular data. Regularization (alpha = 0.001) was applied to control overfitting, particularly due to correlated and encoded features. Furthermore, early stopping was enabled to halt training when performance no longer improves, improving generalization and preventing unnecessary training iterations.

Both models were trained on fully preprocessed data (scaled, encoded, and imputed), ensuring consistency and fairness in comparison. The chosen configurations reflect the underlying characteristics of transportation data, including skewness, outliers, and interdependent variables, and are designed to produce stable, accurate, and interpretable results.

4.3 Model Evaluation

Model performance was evaluated using accuracy, precision, recall, and F1-score to provide a balanced assessment, particularly important given class imbalance in transportation data.

Table 6. Model performance evaluation

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Random Forest (train)	0.986	0.986	0.986	0.986
MLPClassifier (train)	0.926	0.923	0.926	0.923
Random Forest (test)	0.85	0.843	0.85	0.845
MLPClassifier (test)	0.865	0.86	0.865	0.859

The Random Forest model achieves the highest performance on the training data (98.6%), indicating strong learning capability. However, its performance drops to 85.0% on the test set, suggesting some degree of overfitting.

In contrast, the MLPClassifier shows lower training accuracy (92.6%) but achieves slightly better performance on the test data (86.5%). This indicates better generalization to unseen data (Table 6).

Both models perform well on the dominant “Low” traffic class, but performance is weaker for minority classes such as “High” and “Medium”. Notably, the MLPClassifier performs better in identifying critical traffic conditions, which are important for real-world applications.

4.4 Hyperparameter tuning

The Artificial Neural Network (MLPClassifier) was selected over Random Forest based on superior test-set performance, achieving higher accuracy and weighted F1-score. This indicates that the ANN provides better generalization to unseen traffic data, which is critical for real-world deployment.

To further improve performance, hyperparameter tuning was conducted using RandomizedSearchCV with 5-fold cross-validation, optimizing the weighted F1-score to account for class imbalance.

Hyperparameter tuning confirms that the ANN is the preferred model, achieving higher cross-validated performance (weighted F1-score of 0.902). The selected configuration balances model complexity and regularization, making it well suited for transportation classification tasks (Table 7).

Table 7. Hyperparameter search space and optimal parameter values

Hyperparameter	Search Values	Best Hyperparameters (ANN) Value
hidden_layer_sizes	(128, 64), (128, 64, 32)	(128, 64)
activation	relu	relu
alpha	0.0005, 0.001	0.001
learning_rate_init	0.001, 0.0005	0.001
batch_size	32, 64	32

4.5 Model comparison and transportation insight

4.5.1 Final Model Performance

The tuned ANN model achieves good overall performance with an accuracy of 85.7% and a weighted F1-score of 0.85, indicating reliable generalization.

The model performs very well on the dominant "Low" class (F1 ≈ 0.95) with 405 correct predictions, but shows weaker performance on minority classes such as Critical, High, and Medium, particularly in recall (Figure 15). This reflects the impact of class imbalance, where rare traffic conditions are harder to predict.

Best ANN classification report (test):

	precision	recall	f1-score	support
Critical	0.96	0.63	0.76	35
High	0.59	0.64	0.62	59
Low	0.92	0.99	0.95	411
Medium	0.66	0.52	0.58	95
accuracy			0.86	600
macro avg	0.78	0.69	0.73	600
weighted avg	0.85	0.86	0.85	600

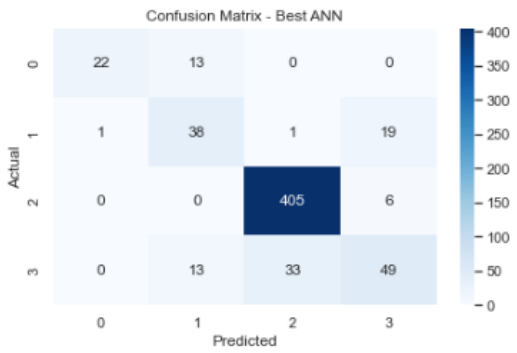


Figure 15. Best ANN (MLPClassifier) classification report (test)

The MLPClassifier (ANN) performed best overall, particularly in terms of test-set generalization. The untuned ANN achieved the highest test accuracy (0.865) and weighed F1-score (0.859), outperforming Random Forest. Although the tuned ANN showed slightly lower performance (accuracy = 0.857), it remains competitive and more robust. In contrast, Random Forest achieved very high training performance but lower test performance, indicating overfitting (Table 8).

Table 8. Model Performance Evaluation (Tuned)

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Best tuned MLPClassifier (test)	0.857	0.851	0.857	0.85

4.5.2 Factors Contributing to Poor Model Performance

Several factors contribute to reduced model performance:

- *Class imbalance*: The dominance of the "Low" traffic class makes it harder for models to accurately predict minority classes such as "High" and "Critical."
- *Overfitting (Random Forest)*: Extremely high training accuracy (0.986) compared to lower test accuracy (0.85) suggests the model memorized training data rather than generalizing.
- *Complex data relationships*: Traffic data contains nonlinear and interdependent patterns (e.g., flow–speed–density), which simpler models may not fully capture.
- *Noise and outliers*: Transportation data often includes extreme events (incidents, congestion spikes), which can affect model stability.
- *Hyperparameter sensitivity (ANN)*: Slight performance drop after tuning indicates that overly constrained parameters can reduce model flexibility.

4.5.3 Implications for Transportation Management Systems

The results have several practical implications:

- *Reliable prediction of traffic states*: The ANN model can better generalize to new data, making it suitable for real-time traffic monitoring systems.
- *Improved incident detection*: Better performance in minority classes (e.g., critical conditions) supports early identification of congestion or disruptions.
- *Decision support*: Accurate classification enables smarter traffic control strategies, such as adaptive signal timing or rerouting.
- *Need for balanced data strategies*: Addressing class imbalance (e.g., resampling or weighting) can further improve system reliability.
- *Model selection matters*: Choosing models that generalize well (like ANN) is more important than those that only perform well on training data.

Overall, ANN provides a more practical and scalable solution for deployment in transportation management systems.

5 Problem 4: Traffic Flow Prediction (Regression)

5.1 Introduction

This problem aims to predict traffic outcomes (e.g., travel time or flow) using sensor data. Two models, Artificial Neural Network (ANN) and Gradient Boosting Regressor are trained and evaluated using RMSE, MAE, and R^2.

Feature importance is analyzed to identify key factors influencing traffic conditions, and the final model is applied to the test dataset to assess its predictive performance.

5.2 Model Training

Two regression models were trained using the processed dataset: an Artificial Neural Network (MLPRegressor) and a Gradient Boosting Regressor. The use of both models enables a structured comparison between a neural network approach capable of capturing complex nonlinear relationships and a boosting-based ensemble method that performs strongly on structured tabular data while providing interpretable feature importance.

5.2.1 Artificial Neural Network

The MLPRegressor was used for its ability to capture complex nonlinear relationships in traffic data, such as interactions between speed, flow, and congestion. A multi-layer architecture (128, 64, 32) with ReLU activation and the Adam optimizer enables the model to learn hierarchical patterns effectively.

To ensure stable training and generalization, a lower learning rate (0.0005) and regularization (alpha = 0.001) were applied. Early stopping was used to prevent overfitting, while a higher number of iterations (max_iter = 1000) ensures convergence. This configuration allows the ANN to model complex traffic dynamics effectively.

5.2.2 Gradient Boosting Regressor

The Gradient Boosting Regressor was used for its strong performance on structured tabular data and its ability to model nonlinear relationships through sequential learning. With 500 estimators and a low learning rate (0.03), the model achieves a good balance between bias and variance.

Regularization parameters such as max_depth = 3, min_samples_split = 5, and min_samples_leaf = 2 help prevent overfitting, while subsample = 0.8 improves generalization. Additionally, the model provides feature importance, making it useful for interpreting key factors affecting traffic prediction.

5.3 Model Evaluation

The regression models were evaluated using RMSE, MAE, and R^2, providing a comprehensive assessment of prediction accuracy, error magnitude, and explained variance.

Gradient Boosting shows slightly lower RMSE and higher R^2, indicating a marginally better fit on training data, while the ANN achieves lower MAE. The MLPRegressor slightly outperforms Gradient Boosting on testing data in terms of RMSE and R^2, indicating better predictive accuracy, while Gradient Boosting achieves a slightly lower MAE (Table 9).

Table 9. Model Evaluation metrics

Model	RMSE	MAE	R^2
MLPRegressor (train)	2.502	1.649	0.974
Gradient Boosting (train)	2.448	1.727	0.975
MLPRegressor (test)	5.533	2.726	0.917
Gradient Boosting (test)	5.587	2.671	0.916

In general, both models perform well, but the MLPRegressor is marginally better overall, as it achieves lower prediction error and higher explanatory power on unseen data. This suggests it is more suitable for deployment in traffic prediction tasks, while Gradient Boosting remains a strong and interpretable alternative.

5.4 Feature Importance

The feature importance analysis shows that navigation_eta_minutes is by far the most important feature, contributing most of the model's predictive power (dominant importance close to 90%). This indicates that estimated travel time is the strongest predictor of actual travel time.

The next most influential features, though significantly less impactful are avg_speed_kph, demand_index and signal_density_per_km. All remaining variables, including precipitation, route characteristics, incidents, and categorical features (e.g., road type, weather), have minimal contribution to the model (Figure 16).

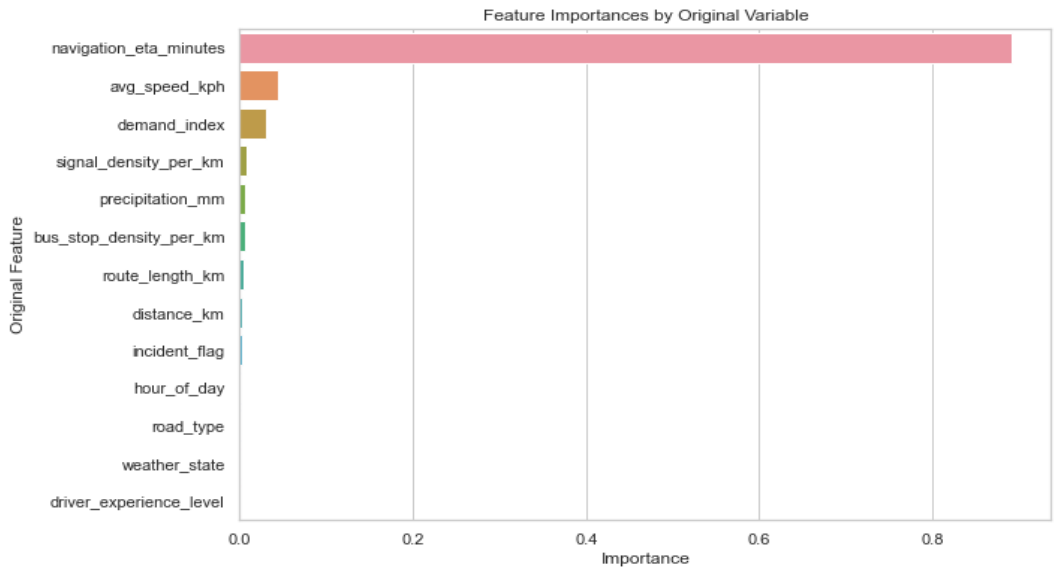


Figure 16. Feature importance from the Gradient Boosting model

5.5 Test Prediction Results

The test predictions demonstrate that the regression model (MLPRegressor) provides reasonably accurate estimates of travel time, with predicted values closely aligned to actual observations in most cases. For example, several predictions (e.g., 41.52 vs 41.27, 18.97 vs 19.52) show minimal error, indicating strong model performance under typical traffic conditions.

However, some discrepancies are observed, particularly for certain instances where the model either overestimates or underestimates travel time (e.g., 21.34 vs 14.85, 26.42 vs 30.14). These deviations suggest that the model may struggle with complex or less frequent traffic scenarios, such as sudden congestion, incidents, or unusual demand patterns (Figure 17).

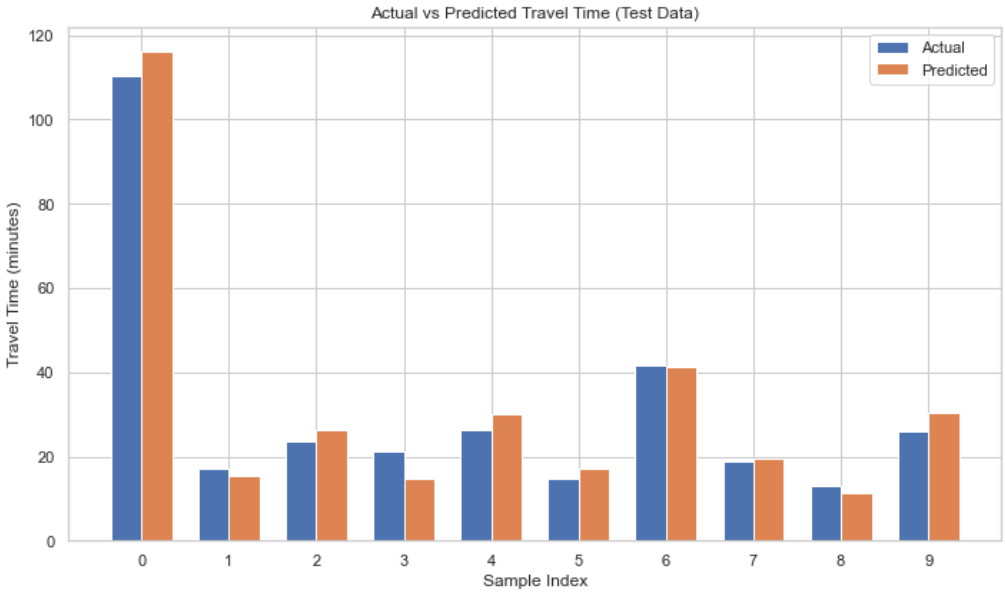


Figure 17. Comparison of actual and predicted travel time values (Sample)

5.6 Model's Suitability for Real-time Forecasting

The regression model (MLPRegressor) is well suited for real-time traffic prediction due to several factors:

- *High predictive accuracy*: The strong R^2 and relatively low error metrics indicate reliable performance for operational use.
- *Generalization capability*: The model maintains stable performance on test data, making it robust for new traffic scenarios.
- *Ability to capture nonlinear relationships*: The ANN effectively models complex interactions between traffic variables, which are common in real-world systems.

However, the model's heavy reliance on key variables such as navigation ETA suggests that its performance depends on the availability and quality of these inputs. Additionally, while the model performs well overall, slight errors in certain predictions indicate that extreme or rare traffic conditions may still be challenging.

Generally, the MLPRegressor provides accurate and reliable predictions, making it suitable for deployment in real-time transportation management systems. It can support applications such as travel time estimation, congestion monitoring, and decision-making, provided that input data remains consistent and reliable.

6 Problem 5: Model Comparison and Interpretation

6.1 Model Comparison

Clustering, classification, and regression differ in their learning type, output, and purpose, and each plays a distinct role in transportation applications.

1) Clustering is an unsupervised learning method that groups data without predefined labels. In transportation, it is used to discover natural traffic patterns, such as identifying states like free flow, moderate congestion, and severe congestion. It helps in understanding system behavior and detecting anomalies when no labeled data is available.
2) Classification is a supervised learning method that assigns observations to predefined categories. In transportation, it is used to predict traffic conditions or risk levels (e.g., low, medium, or critical congestion). It supports real-time decision-making, incident detection, and traffic monitoring systems.
3) Regression is also a supervised method, but it predicts continuous values rather than categories. In transportation, it is used to estimate variables such as travel time, traffic flow, or delay. It is essential for forecasting, route planning, and real-time navigation systems.

Table 10. Key Differences

Aspect	Clustering	Classification	Regression
Learning Type	Unsupervised	Supervised	Supervised
Output	Groups (clusters)	Categories (classes)	Continuous values
Purpose	Pattern discovery	Decision-making	Prediction/forecasting

6.2 Applications for Transportation

Clustering, classification, and regression models play complementary roles in transportation systems, particularly in traffic management, congestion mitigation, and Intelligent Transportation Systems (ITS).

1) Clustering is used to identify natural traffic patterns and operating states (e.g., free flow, moderate congestion, severe congestion). This helps traffic managers understand system behavior, detect anomalies, and monitor network conditions without predefined labels.
2) Classification enables real-time prediction of traffic conditions, such as identifying congestion levels or delay risk categories. This supports rapid decision-making, incident detection, and automated control systems in ITS.

3) Regression provides continuous predictions such as travel time and traffic flow. These predictions are essential for route optimization, congestion forecasting, and real-time traveler information systems.

Table 11. Models in transportation applications (examples)

Model Type	Traffic Management	Congestion Mitigation	Intelligent Transportation Systems (ITS)
Clustering	Identifies traffic patterns and operating states (e.g., free flow vs congestion)	Detects congestion zones and recurring traffic patterns	Supports anomaly detection and traffic state monitoring
Classification	Classifies real-time traffic conditions (e.g., delay risk levels)	Provides early warning of congestion and incidents	Enables automated incident detection and control decisions
Regression	Predicts travel time and traffic flow for planning	Forecasts future congestion and delays	Supports real-time navigation, route optimization, and demand prediction

7 Conclusion

This report developed a complete and structured machine learning workflow for transportation analytics, highlighting the importance of data preprocessing, model selection, and task-specific evaluation. The datasets exhibited real-world challenges such as missing values, skewness, outliers, and multicollinearity, which were effectively addressed using a robust preprocessing pipeline.

The clustering analysis demonstrated that k = 3 provides meaningful and stable traffic regimes, corresponding to free flow, moderate congestion, and severe congestion. In terms of performance, K-Means outperformed hierarchical clustering, achieving a higher silhouette score (0.235 vs 0.215) and a lower Davies–Bouldin index (1.454 vs 1.585), confirming its suitability for traffic state discovery.

For the classification task, the Artificial Neural Network (ANN) achieved the best generalization performance, with test accuracy ≈ 0.865 and weighted F1-score ≈ 0.859, outperforming Random Forest (accuracy ≈ 0.850, F1 ≈ 0.845). Random Forest showed clear signs of overfitting, with training accuracy ≈ 0.986, while ANN maintained more balanced performance. Hyperparameter tuning further validated ANN as the preferred model, achieving a cross-validated F1-score of ≈ 0.902.

In the regression task, both models performed strongly, with the MLPRegressor slightly outperforming Gradient Boosting on test data (RMSE ≈ 5.53 vs 5.59, R^2 ≈ 0.917 vs 0.916), while Gradient Boosting achieved slightly lower MAE (≈ 2.67 vs 2.73). Feature importance analysis revealed that navigation_eta_minutes dominates prediction (~90% importance), with secondary contributions from speed and demand-related variables.

From an application perspective, the results demonstrate that:

- Clustering enables identification of stable traffic regimes for monitoring
- Classification (ANN) provides reliable real-time traffic state prediction
- Regression (MLP/GBR) delivers accurate travel time forecasting

These findings support key transportation applications such as traffic management, congestion mitigation, and intelligent transportation systems (ITS). However, challenges such as class imbalance, model sensitivity, and reliance on dominant features highlight areas for further improvement.

Overall, the report confirms that robust preprocessing combined with appropriate model selection leads to accurate, interpretable, and deployable transportation models, enabling more effective and data-driven traffic management solutions.

8 AI Assistance Statement

AI tools were used in a limited and transparent support role during the preparation of the code and this report. Specifically, AI assistance contributed to code debugging, structuring the workflow into clearly defined exam problems, and refining explanatory text related to standard machine learning procedures. However, all methodological decisions including the selection of preprocessing strategies, model choices, evaluation metrics, and interpretation of results were independently reviewed and validated. The work therefore maintains academic integrity by using AI as a support tool for drafting and organization, rather than as a substitute for analysis or understanding.